

NAME: KARIVUKI DERRICK YADNENG

ADM# 19057

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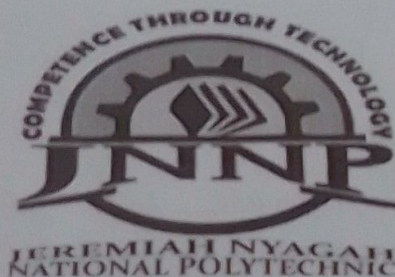
CLASS# EE/245/26

Electrical Engineering (Power Option) Level 6

ENG/OS/PO/CC/02/6

Apply Electrical Principles

Jan - April 2025



JEREMIAH NYAGAH NATIONAL POLYTECHNIC

CONTINUOUS ASSESSMENT TEST
WRITTEN ASSESSMENT

Time: 1 Hour

Total marks: 30

INSTRUCTIONS TO CANDIDATES

1. Answer ALL Questions

Marks for each question are as indicated.

No use of electronic gadgets or any reference material

1. Define the term earthing and explain its main purpose in an electrical installation. (3 marks)

→ Earthing → is the process by which electric current is connected to the ground.
→ H.S. Main purpose is for protection and safety purpose.

2. List three main components of an earthing system and briefly explain their functions. (3 marks)

- ① Earthing continuity conductors
- ② Earthing terminal
- ③ Earthing electrode / protective rod

3. Explain the difference between a TN and a TT earthing system. (4 marks)

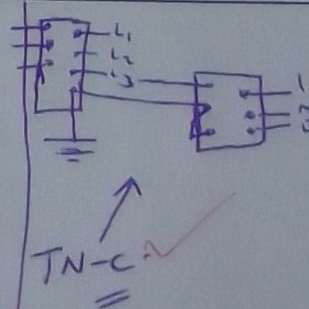
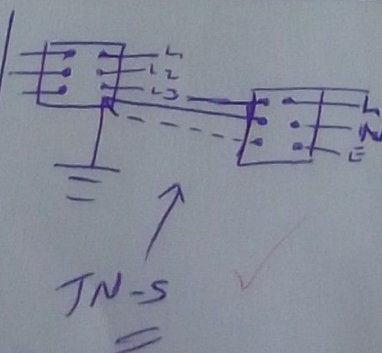
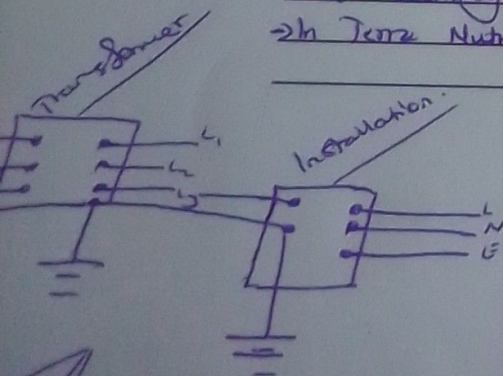
→ TN → is known as Terra Neutral earthing system.
→ The earthing is taken from a Neutral conductor on the transformer and goes to the installation as a separate conductor.
→ TT → is known as Terra Terra earthing system.
→ The earthing is done on both sides. ~~at~~ the earthing is taken from the neutral conductor.

4. Briefly describe the IT earthing system and state one advantage of using it. (3 marks)

~~The system has the conductors ^{insulated} and ^{connected} to earth.~~
~~The system has ^{insulated} earth wire.~~
~~It is easy to implement where safety environment was essential for example in hospital.~~

5. Differentiate between TN-S, TN-C, and TN-C-S earthing systems. (6 marks)

→ ~~Terra Neutral - Combined~~ → ~~Terra Neutral - Combined and Separate~~.
~~Terra Neutral - S~~ → ~~Denotes~~ as Terra Neutral - Separate earthing system.
→ In Terra Neutral separate the current is carried by three conductors where the earthing is taken from the Neutral in the transformer.
→ In Terra Neutral - Combined the current is carried by two conductors where the earthing is taken from the Neutral in the installation.
→ In Terra Neutral Combined & Separate.



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6. State four key factors to consider when selecting a method of earthing for an electrical installation. (4 marks)

- ① The Soil texture.
- ② The type of electrical installation done.
- ③ The cost of earthing.
- ④ The thermal stability.

7. According to IEEE regulations on earthing, state three important guidelines to ensure an effective earthing system. (3 marks)

- ① The Soil texture of a place should be tested.
- ② The earthing conductors should be of good conductivity such as copper.
- ③ The earthing should be done on every installation done with effective earth system according to current used.

8. Why is a low earth resistance important in an earthing system? (4 marks)

- It enhances the conductivity of electric current to the ground.
- Enhance minimizing of electric shock when fault occurs.
- It improves the efficiency of the earthing system.

